

AUTOMATIC METADATA TAGGING IN TEXTUAL LEARNING OBJECT

**A Major Project Report Submitted
In Partial Fulfillment of the Requirements
for the Degree of**

**Bachelor of Technology (B.Tech)
in**

**Information Technology
by**

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to the

**DEPARTMENT OF INFORMATION TECHNOLOGY
DR. APJ ABDUL KALAM TECHNICAL UNIVERSITY
LUCKNOW**

(Formerly Uttar Pradesh Technical University, Lucknow)

May, 2026

DECLARATION

We hereby declare that the work presented in this report, entitled "AUTOMATIC METADATA TAGGING FOR TEXTUAL LEARNING OBJECTS", was carried out by us. We have not submitted the matter embodied in this report for the award of any other degree or diploma of any other University or Institute. We have given due credit to the original authors/sources for all words, ideas, diagrams, graphics, computer programs, experiments, and results that are not our own. We have used quotation marks to identify verbatim sentences and given credit to the original authors/sources. We affirm that no portion of our work is plagiarized, and the experiments and results reported in the report are not manipulated. In the event of a complaint of plagiarism and the manipulation of the experiments and results, we shall be fully responsible and answerable.

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ABSTRACT

Students on modern e-learning platforms often struggle to grasp the correct order for studying complex subjects. Most learning materials lack clear prerequisites or progression paths. As a result, learners navigate content randomly, leading to confusion and gaps in understanding.

This issue matters because self-paced learners rely heavily on digital resources and need structured guidance to learn well. Manually tagging prerequisites is slow, inconsistent, and unworkable for large content libraries. Without automation, keeping a meaningful learning sequence becomes a challenge.

To tackle this problem, the project offers an automated metadata tagging system for textual learning objects. Using natural language processing (NLP) and machine learning (ML), the system identifies key topics, extracts prerequisite knowledge, and organizes concepts into a logical learning hierarchy.

The system creates a personalized and well-structured learning roadmap that guides students' step- by-step through a subject. This increases content accessibility, improves searchability, and makes navigation easier. As a result, learners gain a guided, efficient, and coherent learning experience.

In the future, the system could be enhanced with adaptive learning features that adjust sequences based on individual performance. It may also handle multimedia content, provide real-time feedback, and support cross-domain concept mapping. These improvements can further enhance personalized and smart e-learning.

ACKNOWLEDGEMENT

We are delighted to present the report of our B. Tech project completed during our final year. We would like to express our sincere gratitude to Assistant Professor Dr. Siddhartha Kumar Arjaria, Department of Information Technology, Rajkiya Engineering College, Banda, for their unwavering support and guidance throughout the project. Their diligence, thoroughness, and perseverance were a constant source of motivation for us and played an instrumental role in the successful completion of our project.

We are grateful to our Head of Department of Information Technology, Dr. Vibhash Yadav, for his continuous encouragement and valuable insights throughout the project. We also extend our sincere thanks to all the faculty members of the department for their kind assistance and cooperation during the project development.

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LIST OF ABBREVIATIONS

NLP	Natural Language Processing
ML	Machine Learning
TF-IDF	Term Frequency -Inverse Document Frequency
PDF	Portable Document Format
TXT	Text File
UI	User Interface
JSON	JavaScript Object Notation
LR	Logistic Regression
NB	Naive Bayes

CHAPTER 1

INTRODUCTION

The fast growth of e-learning platforms has made educational content easily accessible to learners. However, students often struggle to find the right learning path for complex subjects because prerequisites are not clearly defined. Learning resources are usually presented without a structured order, which can disrupt understanding. Organizing content manually takes a significant amount of time and can be inconsistent. Therefore, we need smart systems that can automatically analyze learning materials and systematically support learners. This project aims to tackle these issues through automated metadata tagging and the identification of prerequisites.

1.1 IEEE 1484 Learning Object Metadata (LOM) Standard

IEEE 1484 Learning Object Metadata (LOM) is a metadata standard developed by the Institute of Electrical and Electronics Engineers for the description of educational resources in a structured and interoperable manner. It defines a set of metadata elements that describe key features of learning objects such as content, technical form, instructional use, and classification. This standard organizes information in categories such as General, Educational, Technical, and Classification to index, share, retrieve, and reuse e-learning material in an efficient manner across different e-learning platforms.

Within the framework of the proposed system for Automatic Metadata Tagging and Prerequisite Detection, the functionality conforms to the principles of IEEE 1484 LOM. The system automatically extracts key concepts, subject classification, and prerequisite relationships from textual learning resources, matching LOM categories such as Classification, Educational, and Relation. The system produces structured metadata and logical dependencies between concepts to enable smart organization and sequencing of the learning content. The proposed approach can therefore be seen as a step towards the automation of LOM-compliant metadata generation, improving scalability, consistency, and usability in current e-learning environments.

1.2 E-Learning Platforms

The rapid growth of digital education has made a wide range of educational resources available through e-learning platforms. These platforms offer self-paced learning opportunities in many subjects, allowing learners to access content anytime and from anywhere. However, the sheer volume of materials can make it challenging to navigate and understand complex topics. Learners often find it hard to decide which subjects to study first and how different concepts connect. This can result in inefficient and scattered learning experiences.

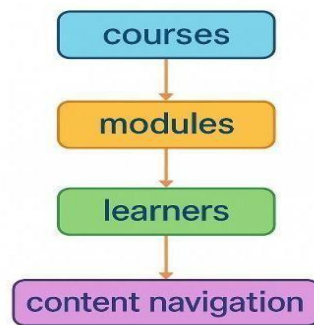


Figure 1.1-Typical E-Learning Platform

1.3 Metadata Tagging in Education

Metadata plays an important role in organizing and managing educational content. Properly tagged metadata, like keywords, learning levels, and topic relationships, improves searchability, navigation, and content discovery. Right now, most e-learning platforms depend on manual tagging, which takes a lot of time, is inconsistent, and hard to scale. Inconsistent metadata often results in poorly structured learning paths. This makes it difficult for learners to progress logically through a subject.

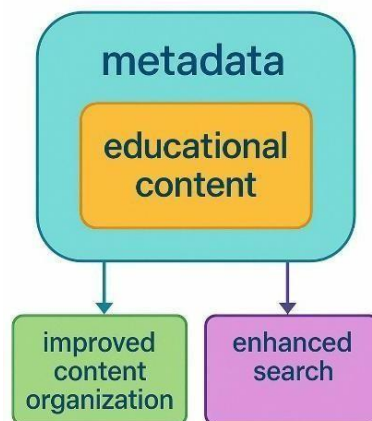


Figure 1.2 - Metadata in Educational Content

1.4 Prerequisite Mapping

A key factor in effective learning is understanding the relationships between topics. Prerequisites ensure that learners gain foundational knowledge before advancing to more complex concepts. This approach helps reduce cognitive overload and improve comprehension. In digital settings, these relationships are often unclear, which forces students to make assumptions or revisit topics frequently.

Automating the identification of prerequisites can significantly improve learning efficiency and content organization. Prerequisite mapping also supports a more personalized and learner-focused approach to education. When prerequisite relationships are clearly defined, learning platforms can guide students through topics in a logical order based on their existing knowledge. This reduces frustration from facing advanced material too soon and helps learners build confidence as they move forward. By facilitating a smooth transition from basic to complex concepts, prerequisite mapping makes learning more intuitive, organized, and effective, especially in self-paced digital environments.

1.5 NLP In Education

NLP provides effective tools to analyze and understand textual learning content. Techniques like concept extraction, keyword identification, and semantic analysis can automatically identify important topics and their connections. By using NLP and machine learning, educational platforms can automate metadata tagging, create structured learning hierarchies, and develop personalized learning pathways without human effort.

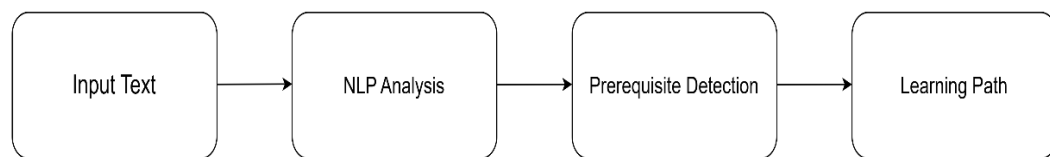


Figure 1.3 NLP-based Workflow

This figure explains how NLP is used to work with educational text in a step-by-step way. First, the system takes raw learning material, such as notes or study content, as input. It then analyzes the text to understand important words, concepts, and their meanings. Based on this understanding, the system identifies which concepts should be learned before others. Finally, useful metadata is created for the content, making learning materials easier to organize, search, and use in digital learning platforms.

1.6 Problem Statement-

To develop an automated text-based system that analyzes educational content to extract key concepts and identify their prerequisite topics, generating a logical learning roadmap for better concept understanding.

1.7 Objectives

The primary objectives of this project are:

- To automate metadata tagging for textual learning objects using NLP and ML.
- To identify key topics and determine prerequisite relationships between them.
- To generate a structured learning roadmap that guides students step-by-step through complex subjects.
- To improve content accessibility, searchability, and overall navigation in e-learning platforms.

CHAPTER 2

LITERATURE REVIEW

2.1. Literature Survey

This project focuses on automatically identifying metadata and prerequisite relationships in textual learning content to support structured and personalized learning. To understand existing techniques and developments in this area, relevant research has been reviewed. These studies explore the use of Natural Language Processing, machine learning, and graph-based models for concept extraction, detecting prerequisites, and generating learning paths. The insights from this literature have helped us understand current challenges, limitations, and best practices. This knowledge provides a solid foundation for designing the proposed system.

Juhi K. Gor & Nidhi A. Doriya (2016) – Generation of a Learning Pathway for Learning Material [1] This paper focuses on generating structured learning pathways using metadata and topic relationships. The authors emphasize pre-study and post-study dependencies to guide learners through content logically. The approach improves content navigation and learning continuity. However, it relies on predefined metadata rather than automated extraction. The study reinforces the importance of sequencing in e-learning environments.

M. B. Rosé et al. (2018) – Automatic Extraction of Prerequisites between Learning Objects [2] This work presents a machine learning approach for automatically extracting prerequisite relationships from textual learning objects. The method relies on feature extraction and classification techniques to identify concept dependencies. It supports adaptive course sequencing and reduces reliance on expert -defined prerequisites. However, the approach may struggle with deep semantic understanding. Despite this, it lays important groundwork for later NLP-based prerequisite research.

Irene Li et al. (2020) – R-VGAE: Unsupervised Prerequisite Chain Learning [3] Irene Li et al. proposed an unsupervised relational variational graph autoencoder to learn prerequisite chains among concepts. The model represented concepts as nodes in a graph and inferred dependencies without labeled data. This approach significantly improved scalability across domains. The work introduced graph-based deep learning into prerequisite modeling.

Irene Li et al. (2021) – Unsupervised Cross-Domain Prerequisite Chain Learning [4] Irene Li et al. extended unsupervised prerequisite learning to cross-domain scenarios using variational graph autoencoders. The method transfers prerequisite relationships from data-rich domains to domains with limited resources. This approach improves learning path generation where labeled data is scarce. However, its effectiveness depends on the similarity between source and target domains. The study contributes to scalable prerequisite modeling across disciplines.

Irene Li et al. (2022) – A Transfer Learning Pipeline for Educational Resource Discovery [5] Irene Li et al. proposed a transfer learning–based pipeline to improve the discovery of high- quality educational resources. The approach leverages pretrained language models to understand semantic relevance between learning materials and user queries. It enhances accessibility to educational content by reducing dependency on manual curation. However, the work primarily focuses on resource discovery rather than explicit prerequisite identification. The study demonstrates the effectiveness of transfer learning in educational NLP applications.

Alireza Ahadi et al. (2022) – Text Mining in Education: A Systematic Review [6] Ahadi et al. provide a comprehensive review of text mining and NLP applications in educational research. The study analyzes trends in concept extraction, assessment, and learning analytics. It highlights research gaps in automated prerequisite detection and learning path generation. The review serves as a valuable reference for identifying future research directions. It confirms the growing relevance of NLP in education.

Abdessamad Chanaa & Nou – Eddine El Faddouli (2023) – Prerequisites-based Course Recommendation [7] The authors propose a prerequisite-aware course recommendation system using Linked Open Data and machine learning techniques. The system extracts concept prerequisites and matches them with metadata to recommend suitable learning content. It improves recommendation accuracy and learner satisfaction. However, the approach relies heavily on structured semantic data sources. The study demonstrates the value of combining metadata and prerequisite modeling.

Wang, H. & Heffernan, N. (2019) – Knowledge Tracing with Deep Learning [8] This study explored deep learning–based knowledge tracing to model student mastery of concepts. It predicts which concepts learners have mastered and which prerequisites are needed. The approach allows for adaptive learning paths. However, it depends on large amounts of student interaction data.

Zhang, Y. et al. (2020) – Graph-based Educational Content Sequencing [9] The authors used graph models to represent dependencies between learning objects and to sequence content logically. It improves content organization and guides learners’ step by step. The model may struggle when the content lacks an explicit relationship.

Chen, L. & Xu, B. (2021) – NLP-based Concept Extraction in E-Learning [10] This research applied NLP techniques such as keyword extraction, named entity recognition, and semantic similarity to identify important concepts in learning materials. It enhances automatic metadata tagging. The limitation is dependency on domain-specific vocabulary for accuracy

Author(s) & Year	Methodology Used	Advantage	Disadvantage
Gor & Doriya (2016)	Metadata-driven learning path generation	Provides a structured learning sequence; improves navigation	Relies on manually created metadata
Rosé et al. (2018)	Machine learning for prerequisite extraction from text	Automates prerequisite identification; supports adaptive learning	Limited deep semantic context handling
Irene Li et al. (2020)	Unsupervised relational variational graph autoencoder (R-VGAE)	Learns prerequisites without labels data; scalable	Depends on the quality of the concept graph
Irene Li et al. (2021)	Cross-domain variational graph autoencoder	Transfers knowledge to data-scarce domains	Performance affected by domain dissimilarity
Irene Li et al. (2022)	Transfer learning with pretrained NLP models	Enhances semantic understanding; improves resource discovery	Does not directly generate prerequisite chains
Ahadi et al. (2022)	Systematic review of NLP and text mining in education	Identifies research trends and gaps	No implementation or experimental model
Chanaa & El Faddouli (2023)	Linked Open Data and machine learning for recommendations	High recommendation accuracy; structured metadata	Reliant on the availability of semantic data

Wang & Heffernan (2019)	Deep learning–based knowledge tracing	Predicts student mastery; supports adaptive learning	Requires large student interaction datasets
Zhang et al. (2020)	Graph-based content sequencing	Improves content organization and learning flow	Struggles with missing or implicit relational data

Table 2.1 Literature Survey Summary

The review of existing literature shows that while significant progress has been made in metadata tagging, prerequisite extraction, and learning path generation, several gaps remain. Early methods relied on manual metadata and predefined paths, which limited scalability and consistency. Many machine learning approaches also lacked deep semantic understanding and the ability to adapt across different domains. Additionally, no current system fully combines automated metadata tagging, prerequisite identification, and structured learning path generation into one process.

This project addresses these gaps by using NLP and machine learning to automatically tag content, map prerequisite relationships, and create personalized learning sequences. It offers a unified and efficient solution that improves both accessibility and learner guidance.

CHAPTER 3

PROPOSED METHODOLOGY

The proposed methodology aims to automate the process of metadata generation and prerequisite identification from textual learning resources. The system is designed to analyze educational content, extract meaningful concepts, and identify logical prerequisite relationships among concepts. By integrating NLP techniques with TF-IDF-based feature extraction and machine learning models, the proposed approach minimizes manual intervention and improves the accuracy and consistency of learning resource organization.

The methodology follows a modular and sequential workflow, starting from user-uploaded learning material and ending with validated prerequisite relationships and structured metadata. Each module performs a specific function and passes processed information to the next stage, ensuring smooth data flow and scalability. The overall workflow of the system is illustrated through a flowchart, which serves as the basis for the methodological design described in this chapter.

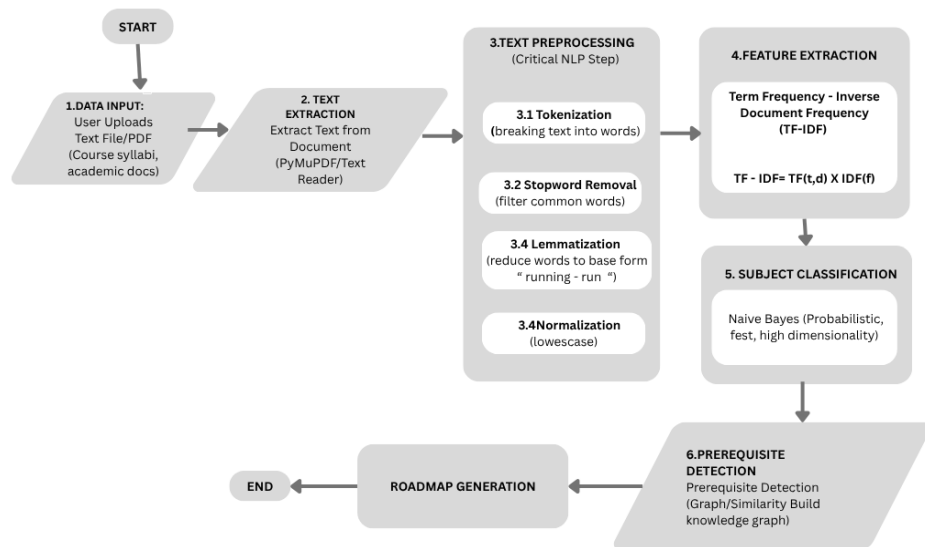


Figure 3.1: It represents the overall workflow of the proposed system.

3.1 User Upload of Learning Content-

The system initiates by enabling users to upload learning materials in either text or PDF format. Supporting multiple file formats ensures that commonly used educational resources can be processed without any additional modification. Once uploaded, these files act as the primary input to the system and are subsequently passed to the text extraction module for further analysis.

3.2 Text Extraction from Documents

Once the learning material is uploaded, the system extracts the textual content using suitable document readers. PDF files are processed with the help of the PyMuPDF library, while plain text files are read using standard text readers. This process converts the content into a raw textual format, ensuring uniformity in processing irrespective of the original file type.

- **PyMuPDF** - It is a lightweight Python library used for efficiently reading and extracting text from PDF documents. It allows fast access to document content while preserving the structure required for further text processing and analysis.

3.3 Text Preprocessing

Text preprocessing is a crucial stage in the proposed methodology, as it prepares the extracted textual content for effective analysis. Raw text obtained from documents often contains inconsistencies, redundant symbols, and words that do not contribute to meaningful understanding. Preprocessing helps in transforming this unstructured data into a clean and standardized form. The preprocessing stage addresses these issues through a sequence of well-defined steps.

3.3.1 Text Normalization-

Text normalization is a crucial preprocessing step that converts raw extracted text into a consistent and standardized format. The primary goal of this process is to ensure that variations in writing style do not affect the analysis of the content. By normalizing the text, the system treats semantically identical words as the same unit, thereby improving the accuracy of further processing stages. The main normalization steps are as follows:

- **Lowercasing of Text**

All characters in the extracted text are converted to lowercase. This prevents the system from treating the same word written in different cases as separate terms. *Example:* “Algorithm” and “algorithm” are treated as the same word.

- **Removal of Punctuation Marks**

Punctuation symbols such as commas, full stops, brackets, and question marks are removed, as they do not add meaningful information for concept extraction.

Example:

The sentence “Machine learning, data science, and AI” is transformed into “Machine learning, data science, and AI” after punctuation removal.

- **Elimination of Special Characters**

Special symbols like @, #, %, and & are removed to reduce noise and maintain clean textual data.

- **Removal of Irrelevant Numbers**

Numerical values are eliminated when they do not contribute to understanding learning concepts, helping to simplify the text for analysis.

- **Whitespace Normalization**

Extra spaces, tabs, and line breaks are replaced with a single space. This improves text readability and ensures proper token separation.

Example:

With the whitespace - "This is an example sentence."

Without whitespace, "This is an example sentence."

Through these normalization steps, the text becomes standardized and free from unnecessary variations. This uniform representation improves the performance of subsequent preprocessing stages and enhances the accuracy of feature extraction techniques such as TF-IDF.

3.3.2 Tokenization-

Tokenization is the process of dividing normalized text into smaller, meaningful units known as tokens.

- The normalized text is split into individual words so that each term can be analyzed separately.
- Tokenization helps the system understand the contribution of each word to the overall meaning of the content.
- It enables accurate computation of word frequency and supports techniques such as TF-IDF.
- Tokenized text forms the basis for keyword extraction and concept identification in later stages.

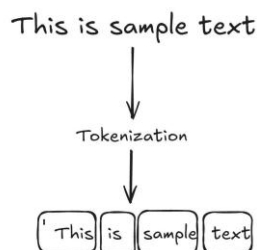


Figure 3.2.2- Tokenization Process

Example:

For the sentence “*machine learning improves educational systems*”, tokenization converts the text into the following tokens: [*"machine", "learning", "improves", "educational", "systems"*].

3.3.3 Stop Words Removal

Stop word removal is performed to eliminate commonly used words that do not contribute significant meaning to the text.

- Stop words such as “*the,*” “*is,*” “*and,*” “*to,*” and “*of*” appear frequently but do not represent important concepts.
- Removing these words helps the system focus on meaningful and domain-specific terms.
- This step reduces noise in the data and improves the effectiveness of feature extraction techniques like TF-IDF.
- Stop Word removal enhances the accuracy of keyword extraction and concept identification.

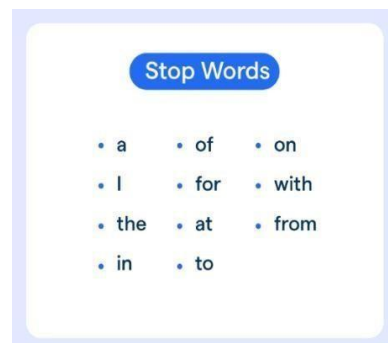


Figure 3.3.3- Stop Word Removal

Example:

For the tokenized sentence [*"the", "system", "is", "designed", "to", "extract", "concepts"*], after stop word removal, the remaining tokens are: [*"system", "designed", "extract", "concepts"*].

3.3.4 Lemmatization

Lemmatization is the process of reducing a word to its base or dictionary form (lemma) while preserving its meaning. Unlike stemming, which may simply cut word endings, lemmatization considers the context and part of speech to ensure that the root word is meaningful.

Purpose: Reduce each word to its base or dictionary form while preserving meaning.

Processing Steps-

1. **Input Tokens:** Take the list of tokens generated after stop word removal.
2. **Convert to Base Form:** Transform each token to its lemma (dictionary form).
3. **Result:** A cleaned set of tokens that reduces vocabulary size but retains semantic meaning.

Benefit:

Lemmatization ensures consistency and helps machine learning models recognize different forms of the same word as a single concept.



Figure 4.3.5- Lemmatization Process

Example: Original Tokens [*'students', 'are', 'running', 'studies', 'better'*] after lemmatization [*'student', 'be', 'run', 'study', 'good'*]

3.3.5 Clean Tokens for ML and Prerequisite Detection

After lemmatization, the text attains a stable and meaningful form in which all words are reduced to their base representations. The resulting tokens are finalized and prepared for direct use in machine learning models and prerequisite analysis, ensuring semantic consistency and relevance.

- Finalized tokens are validated for analytical tasks.
- Only meaningful and relevant concepts are retained.
- Noise and redundancy are eliminated, producing a compact vocabulary.
- Clean tokens are used directly for feature extraction methods such as TF-IDF.
- They support accurate identification of core concepts and prerequisite relationships.

This step ensures that subsequent modules operate on high-quality, ML-ready data, improving the reliability of metadata generation and prerequisite detection.

3.3.6 Preprocessed Text Output

After completing all preprocessing operations, the text is transformed into a clean and well- structured format.

- The processed text is free from noise, inconsistencies, and irrelevant terms.
- All words are standardized, making the data suitable for accurate analysis.
- The cleaned text is forwarded to the feature extraction and concept identification modules.
- Techniques such as TF-IDF are applied to this preprocessed text to compute word importance.
- Proper preprocessing improves the overall efficiency and accuracy of the proposed system.

This step ensures that subsequent modules operate on high-quality input data, leading to more reliable metadata generation and prerequisite detection.

3.4 Feature Extraction and Subject Classification

After preprocessing, the cleaned text data is turned into a number so that machine learning algorithms can work with it. This change is important because machine learning models can't understand raw text directly. The suggested system uses both Term Frequency–Inverse Document Frequency (TF-IDF) to find features and the Multinomial Naïve Bayes algorithm to sort subjects. These methods work together to turn unstructured text into a structured, domain-aware representation that makes it easier to find concepts and prerequisites.

3.4.1 TF-IDF Feature Extraction

The Term Frequency–Inverse Document Frequency (TF-IDF) method is used to figure out how important each term in the document is compared to the whole dataset. It gives more weight to words that are important in a document but not so common in other documents.

Term Frequency (TF) measures how often a term appears in a document and is defined as:

$$TF(t, d) = \frac{\text{Number of occurrences of term } t \text{ in document } d}{\text{Total number of terms in document } d}$$

Inverse Document Frequency (IDF) evaluates how unique or informative a term is across all documents:

$$IDF(t) = \log \left(\frac{N}{1 + df(t)} \right)$$

where N is the total number of documents and $df(t)$ is the number of documents containing the term

t .

The combined **TF-IDF score** is calculated as:

$$TF-IDF(t, d) = TF(t, d) \times IDF(t)$$

The proposed system uses TF-IDF to give more weight to terms that are specific to a subject and informative, and less weight to common words that show up in many documents. This makes a sparse but useful feature vector that accurately shows what the learning material is about. Then, a trained machine learning classifier uses the TF-IDF feature vectors to figure out what the document is about. The system makes sure that all of the processing that comes after it, like subject-aware filtering and concept extraction, stays relevant to the context by figuring out if the content belongs to areas like programming, math, or data structures.

This combination of TF-IDF-based feature extraction and subject classification lets the system go from unstructured text to a structured, domain-aware representation. This is a strong base for accurately identifying concepts and prerequisite detection. The extracted TF-IDF features are subsequently used for subject classification, which directly influences the effectiveness of subject-aware filtering and prerequisite detection in later stages.

3.4.2 Naïve Bayes Classification and Its Implementation

After extracting features, the system uses the Multinomial Naïve Bayes algorithm to sort the document into its correct academic subject. Naïve Bayes is a probabilistic classifier that uses Bayes' Theorem. It works best for text classification tasks because it assumes that features are independent.

The classification is based on the likelihood:

$$P(C | X) = \frac{P(X | C) P(C)}{P(X)}$$

where C represents the subject class, and X represents the TF-IDF feature vector.

In this project, the Multinomial Naïve Bayes model is trained on a dataset with many subjects and their related topics for this project. The model learns the probability distribution of terms for each subject while it is being trained. When a

new document is given, the trained model uses the TF-IDF features from the input text to guess what the most likely subject is.

The implementation uses a Scikit-learn pipeline to combine TF-IDF vectorization and the Naïve Bayes classifier into one process. Also, Grid Search Cross-Validation is used to find the best values for hyperparameters like the smoothing parameter (alpha) and the n-gram range, which makes classification work better.

The predicted subject is very important for the next steps in the system. It lets you filter by subject, so only the most relevant ideas are kept for keyword extraction and prerequisite detection. This greatly improves the system's overall accuracy and reliability in context.

3.5 Subject - Aware Filtering

After subject classification, subject-aware filtering is applied to refine the extracted features by retaining only domain-relevant terms. This step removes unrelated or generic words that do not contribute meaningfully to the identified academic subject.

- Retains only subject-specific vocabulary.
- Eliminates irrelevant and noisy terms.
- Improves the accuracy of concept extraction.
- Reduces feature space and computational overhead.
- Ensures contextual consistency for prerequisite detection.

3.6 Concept and Keyword Extraction

After subject-aware filtering, the system extracts the most relevant concepts and keywords from the learning material. This process identifies terms that represent the core topics, removes generic or irrelevant words, and produces a concise set of concepts. These concepts are then used for analyzing prerequisite relationships, ensuring accurate mapping of dependencies between knowledge components.

- Selects essential concepts representing the document's main ideas.
- Extracts meaningful keywords based on importance and context.
- Eliminates remaining non-informative terms.
- Prepares concept data for prerequisite detection.

3.7 Learning Path

In this step, the system identifies dependencies between concepts to determine which topics need to be understood before others. Using the extracted keywords and concepts, it analyzes relationships and builds a prerequisite map. This ensures learners follow a logical sequence when studying, improving comprehension and learning efficiency.

- Determines which concepts depend on others.

- Uses extracted keywords and concepts to analyze dependencies.
- Builds a prerequisite map for structured learning.
- Supports effective sequencing of learning materials.

3.8 Validated Prerequisite Relationship and Metadata

After identifying prerequisite relationships, the system validates them to ensure they accurately represent the logical dependencies between concepts. This step checks for consistency and correctness, confirming that each prerequisite reflects a true learning progression. At the same time, metadata such as concept importance, frequency, and contextual relevance is attached to each concept. This enriched information helps organize the learning material effectively and supports adaptive learning by highlighting critical concepts and their interconnections.

3.9 Implementation Overview

This chapter shows how to put the proposed system for Automatic Metadata Tagging and Prerequisite Detection in Textual Learning Objects into practice and test it. NLP and ML techniques are used in the system to analyze educational documents and find useful information.

The system looks at text to find important keywords, make metadata, and figure out how concepts are related to each other. It also helps users create structured output and has an interactive interface. Python was used to make the program, along with NLP libraries like NLTK and Scikit-learn. This chapter discusses the tools used, how the system worked, the test results, and how well it performed overall.

Component	Specifications/Tool
Programming Language	Python
NLP Libraries	NLTK, spaCy
ML Libraries	Scikit-learn
Text Extraction	PyMuPDF
Feature Extraction	TF-IDF
Classifier	Naïve Bayes
Interface Framework	Streamlit
Data Handling	Pandas, NumPy
Output Format	JSON

Table 3.1 Software Environment and Tools

The system follows a structured workflow that converts raw input documents into meaningful metadata and learning insights.

3.10 COMPARISON BETWEEN THE EXISTING SYSTEM AND THE PROPOSED SYSTEM

Simple readability metrics like sentence length and word count are what traditional systems use. These methods don't really show what educational content means or how deep it is. They also can't tell how topics are related to each other.

The proposed system addresses these problems by using NLP and ML to better analyse text. It focuses on finding useful concepts, figuring out how they are related, and making structured metadata.

Feature	Existing System	Proposed System
Analysis Method	Basic readability formulas	NLP + ML-Based analysis
Semantic Understanding	Not available	Extract concepts and relationships
Keyword Extraction	Manual or limited	Automated using TF-IDF
Prerequisite Detection	Not supported	Automatically identifies dependencies
Scalability	Limited	Supports multiple documents
Output	Unstructured	Structured metadata (JSON)
User Interface	Minimal	Interactive Streamlit interface

Table 3.2 Comparison between systems

3.11 FUNCTIONAL OUTPUT AND EVALUATION

Several instructional materials from computer science fields like programming, data structures, and operating systems were used to test the system. Every document underwent processing to extract c concepts and keywords, produce metadata, and determine the necessary connections.

Metric	Value
Extracted Keywords	Variables, loops, functions
Subject	Programming
Prerequisite Relationships	Data types – variables – functions
Output Format	JSON

Table 3.3 System Output Summary

The system produced useful and organized outputs that can help students learn about topics in the right order.

3.12 BATCH PROCESSING AND RECOMMENDATION

Batch processing is supported by the system, enabling the simultaneous analysis of several documents. When comparing various educational resources, this function is helpful. Following processing, the system contrasts the concepts that were extracted, determines the degree of intricacy, and suggests appropriate documents.

File Name	Keywords Extracted	Difficulty Level
Doc1.pdf	Variables, loops	Easy
Doc2.pdf	Recursion	Medium
Doc3.pdf	Algorithms, optimization	Hard

Table 3.4 Document Recommendation

3.13 SYSTEM PERFORMANCE

From an implementation standpoint, the system demonstrates:

- Efficient text processing using optimized libraries
- Accurate classification using Naïve Bayes
- Improved keyword extraction through TF-IDF
- Fast execution suitable for real-time applications

The modular design ensures flexibility and scalability for future improvements.

3.14 USER INTERFACE OUTPUT

The user interface is developed using Streamlit, providing an interactive platform for users to upload and analyze documents.

Key features include:

- File upload (PDF/TXT)
- Display of extracted text preview
- Visualization of detected prerequisites
- Concept relationship graph generation
- Structured roadmap display

The interface is designed to be intuitive, enabling both technical and non-technical users to operate the system efficiently.

3.15 COMPARATIVE ANALYSIS OF CLASSIFICATION ALGORITHMS

To evaluate the effectiveness of the chosen classification approach, multiple machine learning algorithms were considered for subject classification, including K-Nearest Neighbors (KNN), Support Vector Machine (SVM), and Multinomial Naïve Bayes. These algorithms were analyzed based on their suitability for text classification tasks, computational efficiency, and performance with high-dimensional sparse data such as TF-IDF vectors. Text classification problems typically involve a large number of features with relatively fewer training samples. Therefore, selecting an appropriate classifier is crucial for achieving accurate and efficient results.

Algorithm	Multinomial Naive Bayes (MNB)	K-Nearest Neighbors (KNN)	Random Forest	Support Vector Machine (SVM)
Type	Probabilistic	Instance-based	Ensemble (Trees)	Margin-based
Working Principle	Uses probability of words given class	Finds nearest neighbors using distance	Combines multiple decision trees	Finds optimal separating hyperplane
Best for Text Data?	Excellent	Poor	Moderate	Good
Handling High Dimensions	Very good	Very poor	Average	Good
Speed (Training)	Very fast	Slow (stores all data)	Moderate	Moderate
Speed (Prediction)	Very fast	Very slow	Moderate	Fast
Memory Usage	Low	Very high	Medium	Medium
Works with Sparse Data (TF-IDF)	Excellent	Poor	Not ideal	Good
Overfitting Risk	Low	Medium	High (if not tuned)	Medium
Accuracy (Text Classification)	High	Low	Medium	High

3.6 Comparison of algorithms

Justification for Selecting Multinomial Naïve Bayes

Based on the comparative analysis, the Multinomial Naïve Bayes classifier was selected for the proposed system due to the following reasons:

- It performs exceptionally well for **text classification problems**, especially with TF-IDF features.
- It is **computationally efficient**, making it suitable for real-time processing.
- It handles **high-dimensional sparse data** effectively.
- It requires **less training data** compared to other algorithms.
- It integrates seamlessly with the Scikit-learn pipeline used in the implementation.

In the proposed system, the Multinomial Naïve Bayes classifier is trained on TF-IDF feature vectors derived from subject-specific datasets. The trained model predicts the subject of the input document, which is further used for subject-aware filtering and prerequisite detection.

The implementation and assessment of the suggested system were covered in this chapter. The solution effectively combines machine learning and natural language processing methods to automate prerequisite detection and metadata tagging.

The outcomes demonstrate the system's ability to efficiently arrange instructional information and produce insightful conclusions from textual input. For contemporary e-learning contexts, it offers a scalable and user-friendly solution.

CHAPTER 4

RESULTS AND ANALYSIS

This chapter presents the experimental results obtained from the implementation of the proposed system for automatic metadata tagging and prerequisite detection. It evaluates the system's performance in terms of concept extraction, subject classification, and prerequisite identification. The analysis also highlights the effectiveness of the applied Natural Language Processing (NLP) and Machine Learning (ML) techniques, along with the limitations observed during testing.

4.1 Experimental Setup

This section explains how the proposed system was set up and tested in order to evaluate its performance in automatic metadata tagging and prerequisite detection.

For experimentation, a structured dataset containing more than 400 subject–topic mappings was used. The dataset covers multiple areas of Computer Science such as Data Structures, Algorithms, Database Systems, Artificial Intelligence, and Programming. These areas were selected to ensure that the system is tested on a wide range of concepts, from basic to more advanced topics. This helps in checking whether the system can work effectively across different domains instead of being limited to a specific subject. Each topic in the dataset is linked with its corresponding prerequisite topics, which serve as a reference during evaluation.

The system was implemented using Python, as it provides a simple and efficient environment for handling text data and building machine learning models. For the machine learning part, the Scikit-learn library was used. It was mainly used to apply the Multinomial Naïve Bayes algorithm and to perform TF-IDF vectorization. TF-IDF helps in identifying the importance of words in a document by giving higher weight to meaningful and less frequent terms, while reducing the importance of commonly used words. This allows the system to focus on the most relevant information present in the text.

The Multinomial Naïve Bayes algorithm was chosen for classification because it works well with text data and is computationally efficient. It uses probability-based learning to classify the given content into the most appropriate subject category. For handling input documents, PyMuPDF was used to extract text from PDF files accurately. In addition, Streamlit was used to build a simple interface so that users can upload documents and easily view the results generated by the system.

The working of the system follows a clear sequence of steps. First, the user uploads a PDF or text file as input. The system then extracts the text from the document. After extraction, the text is cleaned and prepared through preprocessing steps such as removing stopwords, converting text to lowercase, and breaking the text into smaller units (tokens). This step ensures that only useful information is passed to the model.

Next, TF-IDF is applied to convert the processed text into numerical form, which can be understood by the machine learning model. These numerical features are then given to the Naïve Bayes classifier, which predicts the subject category of the input content. Once the subject and key concepts are identified, the system analyzes the relationships between topics to determine prerequisite concepts. Finally, the system produces the output, which includes the classified subject and the list of prerequisite topics.

To evaluate the performance of the system, the generated results were compared with a predefined ground truth dataset. This comparison helps in measuring how closely the system's predictions match the actual prerequisite relationships. Standard evaluation measures such as accuracy, precision, recall, and F1-score can be used to assess the performance in a balanced way.

4.2 Output Results

The system processes the uploaded PDF or text documents and generates the following outputs:

- Predicted Subject of the document
- Extracted prerequisites (keywords/concepts)
- Concept relationships (if detected)
- Study roadmap with definitions

Sample Output

For an input document related to Artificial Neural Networks, the system produced:

- **Predicted Subject:** Machine Learning
- **Extracted Concepts:** Datasets, Precision, Recall, Confusion Matrix, Gradient Descent, Linear Regression, KNN, SVM
- **Study Roadmap:** Definitions and explanations for each concept

The system successfully identified relevant machine learning concepts from the document, demonstrating its ability to process unstructured textual data.

AUTOMATIC METADATA TAGGING IN TEXTUAL LEARNING OBJECTS

Analyze History

Upload PDF/TXT



Drag and drop file here

Limit 200MB per file • PDF, TXT

Browse files



Applied_Artificial_Neural_Networks.pdf 8.4MB



Analyze

Step-by-Step Learning Order ↔

Agent → Environment → Actions → Variables → Domains → Constraints → Resolution → Planning
→ Regression → Robotics → Sensors → Inference → Reasoning → Learning → Classification →
Clustering

Concept Relationships

- ◆ Agent is required to understand Environment
- ◆ Environment is required to understand Actions
- ◆ Actions is required to understand Variables
- ◆ Variables is required to understand Domains
- ◆ Domains is required to understand Constraints
- ◆ Constraints is required to understand Resolution
- ◆ Resolution is required to understand Planning

The above screenshot represents the output generated by the Automatic Metadata Tagging system after successfully analyzing the uploaded document. Once the file is provided, the system processes the text using natural language techniques to identify the most important concepts present in the content. Instead of displaying these concepts in a random manner, the system intelligently organizes them into a step-by-step learning order. This sequence is designed in such a way that each topic builds on the previous one, making it easier for a learner to understand the subject in a logical and structured way s.

For example, basic concepts like “Agent” appear before more advanced topics, ensuring that foundational knowledge is developed first. This approach mimics the way a teacher would naturally explain a subject, starting from simple ideas and gradually moving towards more complex ones.

In addition to the learning sequence, the system also provides concept relationships. These relationships clearly show how one concept is connected to another and why a particular order is followed. It highlights dependencies between topics, helping users understand that certain concepts must be learned before others. This not only improves clarity but also avoids confusion during the learning process.

Deploy

Study Roadmap

agent

Definition:

An agent is an entity in artificial intelligence that perceives its environment and takes actions to achieve specific goals.

environment

actions

Definition:

Actions are the operations or decisions taken by an agent in response to its environment in a computational system.

variables

domains

Definition:

A domain refers to a specific area of knowledge or application in which a problem or system operates.

constraints

inference

resolution

Study Roadmap

big o

Definition:

Big O notation is a mathematical notation that describes the approximate size of a function on a domain. Big O is a member of a family of notations invented by the German mathematicians Paul Bachmann and Edmund Landau and expanded by others, collectively called Bachmann–Landau notation.

[Read more on Wikipedia](#)

time complexity

Definition:

In theoretical computer science, the time complexity is the computational complexity that describes the amount of computer time it takes to run an algorithm. Time complexity is commonly estimated by counting the number of elementary operations performed by the algorithm, supposing that each elementary operation takes a fixed amount of time to perform.

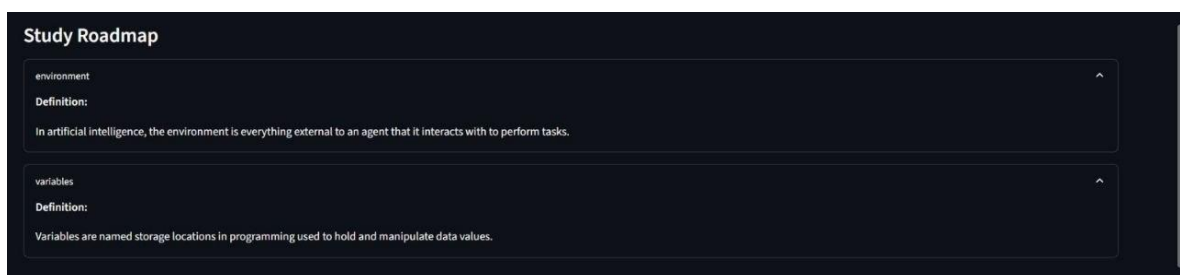
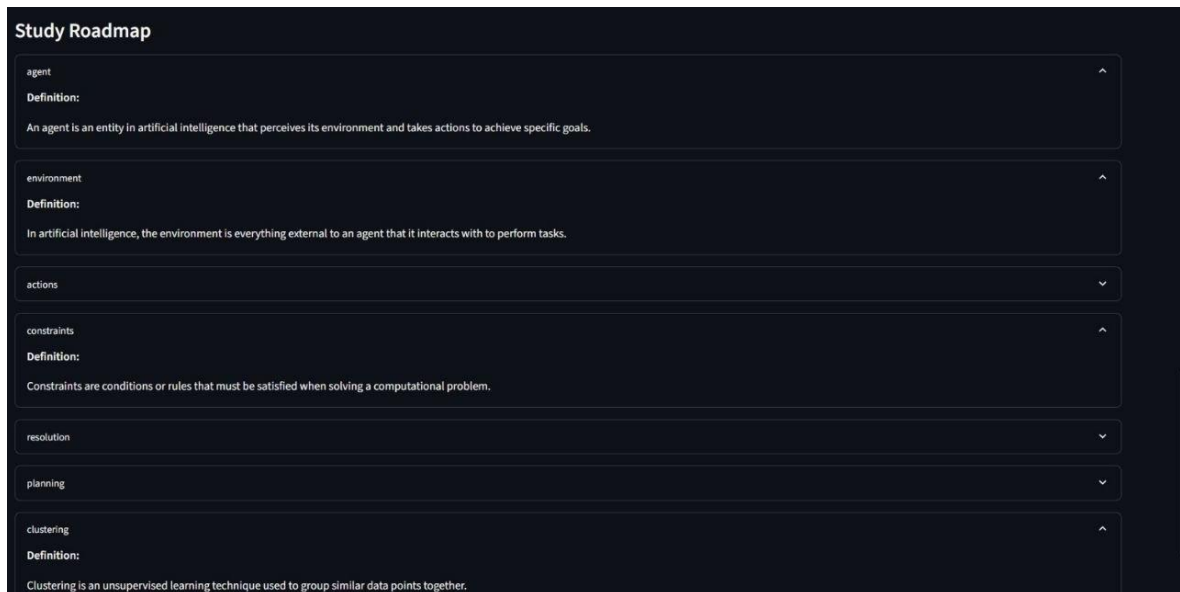
[Read more on Wikipedia](#)

space complexity

Definition:

In computational complexity theory, a complexity class is a set of computational problems "of related resource-based complexity". The two most commonly analyzed resources are time and memory.

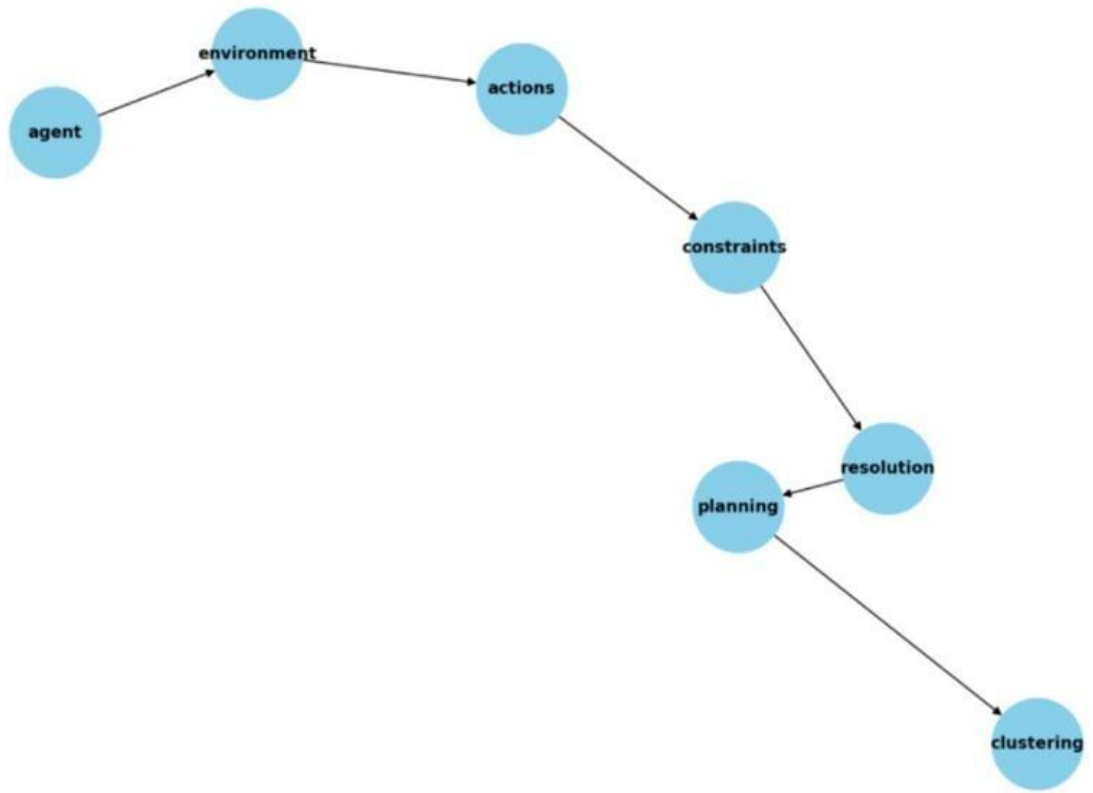
[Read more on Wikipedia](#)



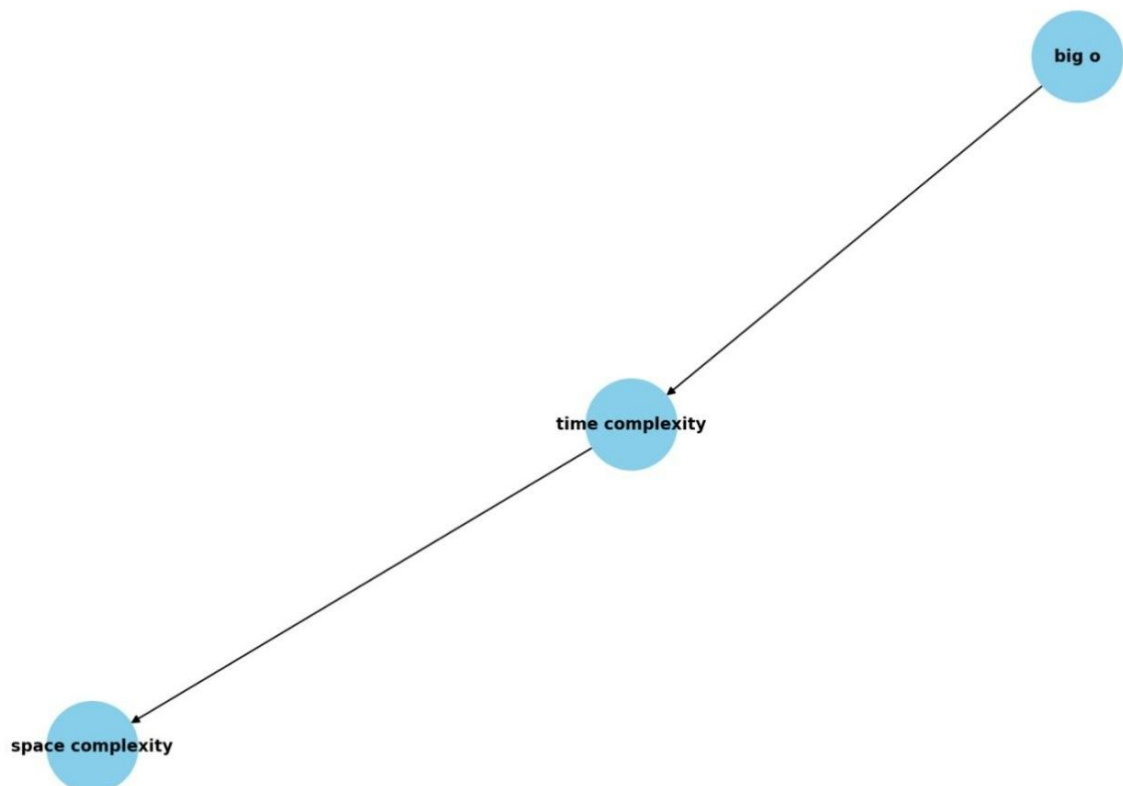
The above output shows the “Study Roadmap” section generated by the system after processing the input document. In this part, all the identified concepts are presented in an organized and user-friendly format, where each concept is displayed along with its definition. The roadmap is structured in a way that allows users to explore each topic individually, making the learning process more interactive and easy to follow.

Each concept, such as “Agent,” “Environment,” “Actions,” and others, is shown as a separate expandable section. When a user selects a particular concept, its definition is displayed, providing a clear understanding of that topic. These definitions are either derived from the processed content or matched from the predefined dataset, ensuring that the information remains relevant and accurate.

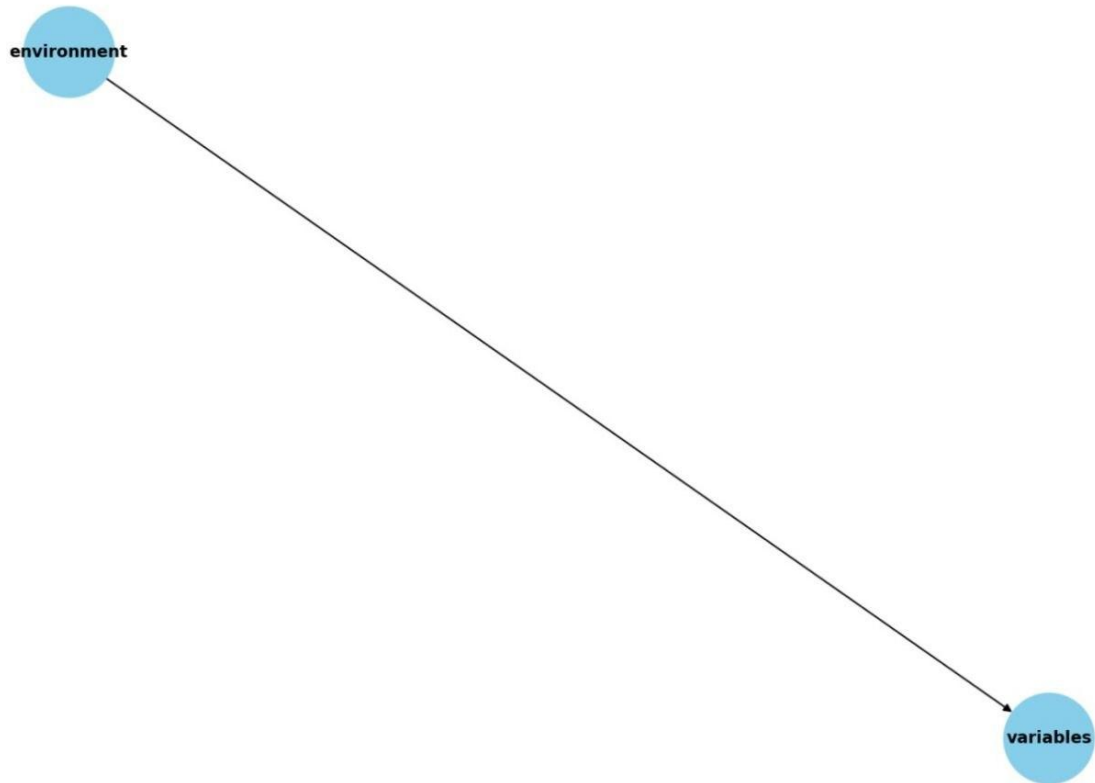
Prerequisite Learning Graph



Prerequisite Learning Graph



Prerequisite Learning Graph



The above screenshot represents the prerequisite learning graph generated by the system, which visually shows how different concepts are connected to each other based on their dependencies. Each node in the graph represents a concept extracted from the document, such as Agent, Environment, Actions, Constraints, and so on. The arrows between the nodes indicate the direction of learning, meaning one concept should be understood before moving on to the next.

For example, the graph starts with Agent, which is connected to Environment, followed by Actions, and then moves towards more advanced concepts like Constraints, Resolution, and Planning. This structure clearly shows that the system does not treat concepts independently, but understands their relationships and arranges them in a meaningful order.

The graph is created by analyzing the extracted keywords and mapping them with predefined dependencies stored in the dataset. These dependencies are then used to build a directed structure, similar to a knowledge graph, where each connection represents a prerequisite relationship.

This visual representation makes it easier for users to understand the flow of learning at a glance. Instead of reading a list of topics, users can see how concepts are linked and how knowledge progresses step by step. Overall, this graph enhances the usability of the system by providing a clear and intuitive understanding of concept dependencies and learning pathways.

CHAPTER 5

CONCLUSION AND FUTURE SCOPE

5.1 CONCLUSION

The rapid growth of digital education has made organizing large volumes of learning content increasingly challenging, often leaving learners confused due to unclear topic sequences and prerequisite relationships. Traditional methods such as manual categorization and metadata tagging are time-consuming, inconsistent, and difficult to scale. To address this, this study introduces an automated system that leverages machine learning and natural language processing (NLP) to extract key concepts, generate metadata, classify subjects, and identify prerequisite relationships. By following a structured pipeline—including text extraction, preprocessing, TF-IDF-based feature representation, classification, and concept extraction—the system ensures improved accuracy, consistency, and efficiency compared to manual approaches. A key contribution of this work is the automatic generation of structured learning pathways that guide learners from foundational to advanced topics, reducing confusion and enhancing comprehension. Combined with a user-friendly interface, the system allows both learners and educators to easily input materials and explore concept relationships interactively. Overall, this project provides a scalable and intelligent solution for organizing educational resources, demonstrating how advanced technologies can support more structured, guided, and personalized learning experiences in modern e-learning environments.

5.2 Future scope

Although the existing automatic metadata tagging and prerequisite detection system yields excellent results, there are a number of areas where it may be improved to boost its effectiveness, precision, and practicality. The system can be developed into a more reliable and intelligent solution for contemporary e-learning environments by adding cutting-edge technology and enhancing its capabilities.

5.2.1 Improved Accuracy using Advanced Models

For categorization and concept extraction, the current system makes use of conventional machine learning approaches. These techniques work well, but they might not adequately capture the intricate semantic links seen in instructional materials.

To enhance contextual awareness in the future, sophisticated deep learning models like transformer-based architectures can be incorporated.

These models can improve the overall accuracy of requirement detection and metadata tagging, as well as better understand sentence structure and uncover hidden connections between concepts.

5.2.2 Handling Large-Scale Data

Currently, the system can only handle a certain number of papers at once. Making the system capable of effectively managing massive datasets is crucial as the amount of digital learning resources keeps increasing.

In order to guarantee quicker processing and scalability, future enhancements can include optimization strategies, parallel processing, and cloud-based deployment. This will enable the technology to be used in huge e-learning platforms, digital libraries, and universities.

5.2.3 Enhanced Concept Relationship Detection

Currently, prerequisite relationships are identified using simple dependency analysis and extracted keywords. Deeper conceptual dependencies might not always be captured by this method, though.

To enhance relationship recognition, future research can concentrate on combining knowledge graphs, graph-based algorithms, and semantic similarity metrics. By using these methods, the system will be able to find more precise and significant links between ideas, leading to more organized learning pathways.

5.2.4 User-Friendly Interface Improvements

The present system offers an easy-to-use interface for document analysis and uploading. However, adding sophisticated visualization tools, interactive dashboards, and thorough findings explanations can improve the user experience even further. Features like step-by-step learning paths and graphical depictions of prerequisite linkages can improve the system's usability and intuitiveness for both teachers and students.

5.2.5 Real-Time Processing and Faster Response

Data is processed by the current system once the upload and analysis cycle is over. As soon as the content is submitted, real-time processing capabilities may be added in the future to deliver immediate results. This will increase the system's responsiveness and make it more appropriate for real-time applications, where users anticipate prompt feedback and insights.

5.2.6 Integration with Educational Platforms

By combining it with current Learning Management Systems (LMS) and online learning environments, the system can be expanded. This interface will enable the direct application of prerequisite mapping and automated metadata tagging in digital learning settings. This will increase overall learning efficiency by providing students with guided learning routes and structured content without the need for manual involvement.

5.2.7 Extension to Multiple Domains

Currently, the system is mostly concentrated on particular kinds of instructional content. It can be expanded in the future to accommodate a variety of academic fields, including professional courses, science, mathematics, and the humanities.

To increase the system's adaptability and usability, it will be necessary to train it on a variety of datasets and modify it to handle terms unique to a given area.

5.2.8 Support for Different File Formats

The system currently supports text and PDF files. Support for more file formats, such as Word documents, presentations, and web-based material, may be added in the future.

As a result, the system will be more adaptable and able to manage a greater variety of educational material.

FINAL REMARKS

This research shows how machine learning (ML) and natural language processing (NLP) techniques. It can be used practically to improve the structure of educational content.

The technology transforms unstructured textual material into useful and organized information by automating precondition detection and metadata tagging. The suggested method reduces confusion and improves overall learning effectiveness by assisting students in comprehending subjects in a logical order.

Additionally, it reduces human labor, increasing the system's scalability and suitability for massive amounts of educational resources. The implementation further demonstrates the usefulness of such a system in actual e-learning settings.

Despite the system's good performance, accuracy, and sophisticated feature integration should yet be improved. The system may become more sophisticated and flexible in the future to accommodate various kinds of instructional materials. To sum up, the project offers an easy-to-use yet efficient solution for material organizing and structured learning. It has the potential to significantly improve digital education systems with more development.

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